CSE306: Geometry Processing Project Report

Nikola Dobricic

June 19, 2025

1 Introduction

This report details the implementation of a series of projects for the CSE306 Computer Graphics course. The project is divided into three main parts: building Voronoï and Power diagrams, solving a semi-discrete optimal transport problem, and developing a free-surface fluid simulator. Each lab builds upon the previous one, with implementations based on the concepts and algorithms described in the course lecture notes [1].

2 Lab 6: Voronoï Diagrams via Polygon Clipping

The first stage of the project was to compute 2D Voronoï diagrams. A Voronoï diagram of a set of sites is a partition of the space into cells, where each cell contains the points closer to its site than to any other.

2.1 Implementation: Voronoï Parallel Linear Enumeration

The chosen method was the Voronoï Parallel Linear Enumeration algorithm. This algorithm computes each Voronoï cell independently. For each site P_i, its cell is determined by starting with a large bounding polygon and iteratively clipping it by the half-spaces defined by the perpendicular bisectors between P_i and all other sites P_j.

The core of this process is the Sutherland-Hodgman algorithm, which was implemented to perform the clipping of a polygon against a line. The naïve version of this algorithm has a time complexity of $O(N^2)$, but was implemented in parallel using OpenMP to improve performance. Figure 1 shows a result from this implementation.

3 Lab 7: Power Diagrams and Optimal Transport

The second lab extended the functionality to Power diagrams and used them to solve a semi-discrete optimal transport problem.

3.1 Power Diagrams

A Power diagram is a generalization of a Voronoï diagram where each site P_i has an associated weight w_i. The cells are defined based on the power distance, given by $\|P - P_i\|^2 - w_i$. This modification causes the cell boundaries to shift, allowing for explicit control over cell sizes. The implementation was a direct extension of the Voronoï code, requiring only a change in the calculation of the clipping line for the Sutherland-Hodgman algorithm. Figure 2 shows a power diagram for sites arranged in a ring.

3.2 Semi-Discrete Optimal Transport

Optimal transport provides a method to find a set of weights for a power diagram such that the area of each cell matches a predefined target mass. This is formulated as a convex optimization problem.

The ‘libLBFGS‘ library was integrated to solve this problem by maximizing a concave functional whose gradient is the difference between the current cell areas and the target areas. The implementation involved writing an ‘evaluate‘ function that computes the power diagram for a given set of weights, calculates the objective function and its gradient, and passes them to the L-BFGS solver. Figure 3 demonstrates the result, where cell sizes are adapted to a target Gaussian density.

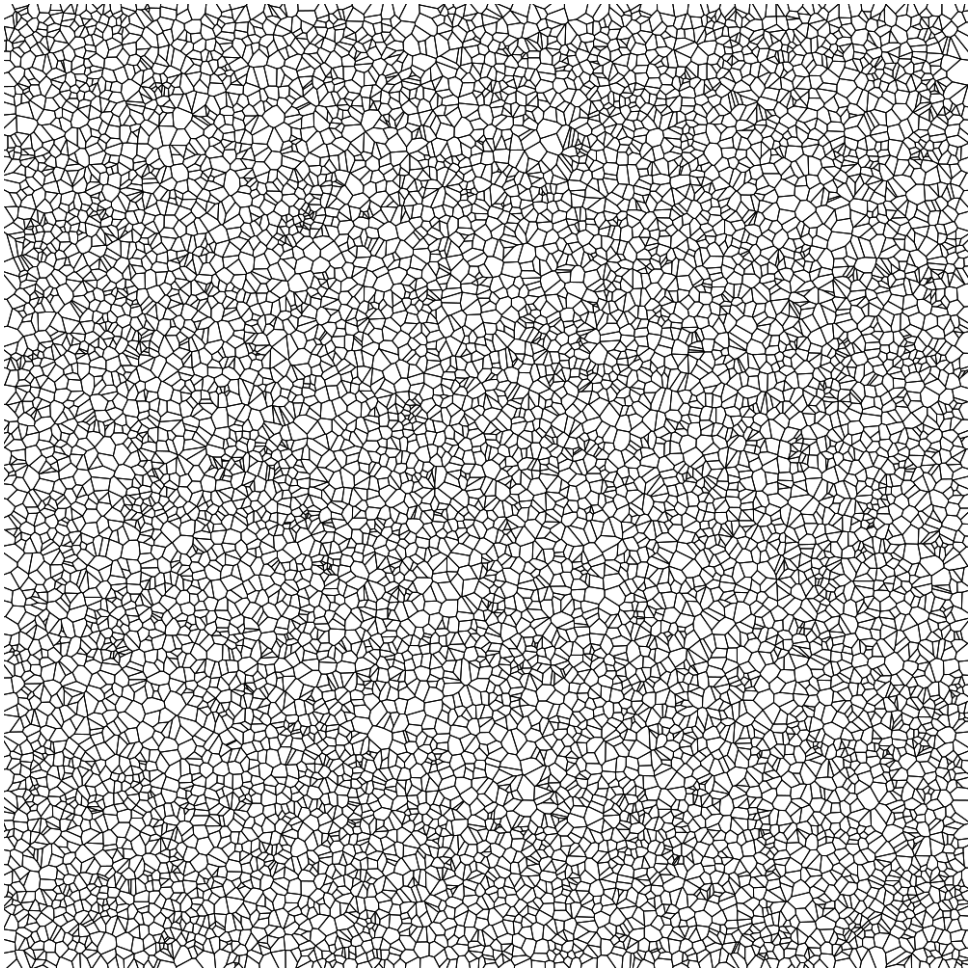


Figure 1: A Voronoï diagram generated for 10,000 random sites using the implemented $O(N^2)$ parallel linear enumeration algorithm.

4 Lab 8: Optimal Transport Fluid Simulation

The final part of the project was to build a 2D free-surface fluid simulator using the optimal transport framework. This provides an elegant, grid-free approach to enforcing the incompressibility of the fluid.

4.1 Methodology

The fluid is modeled as a collection of particles. Incompressibility is enforced by ensuring that the Laguerre cell of each fluid particle maintains a constant volume throughout the simulation. This is achieved by solving a partial semi-discrete optimal transport problem at each time step.

A key challenge is handling the free surface. This is addressed by introducing a single "air" weight, w_{air}, for the entire domain not occupied by fluid. The power cell of each fluid particle P_i is then not only clipped by the bisectors of other fluid particles but also by a disk of radius $R_i = \sqrt{w_i - w_{air}}$. This elegantly models the fluid-air interface.

The dynamics of the particles are driven by gravity and a spring force that pulls each particle towards the centroid of its Laguerre cell. This spring force is the mechanism that drives the fluid towards an incompressible state.

4.2 Simulation Results

The resulting simulator can capture dynamic fluid behaviors like splashing and settling. The simulation frames are rendered by drawing the Laguerre cells of the fluid particles. Figure 4 presents snapshots from a simulation of a collapsing disk of fluid under gravity.

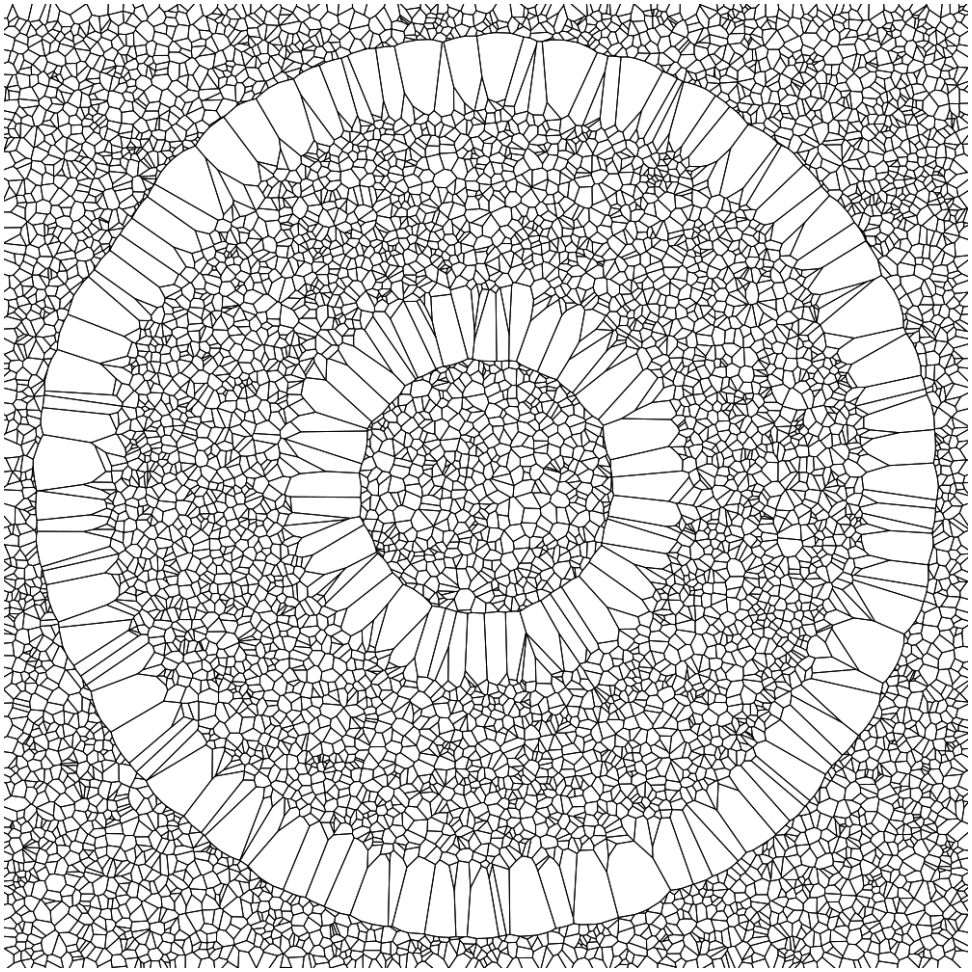


Figure 2: A Power Diagram for 10,000 sites. The weights assigned to the sites alter the cell shapes compared to a standard Voronoï diagram.

5 Conclusion

The project successfully progressed from implementing fundamental structures like Voronoï diagrams to a modern fluid simulation technique based on optimal transport.

References

[1] Course lecture notes, Chapters 4 & 5. *Provided for CSE306.*

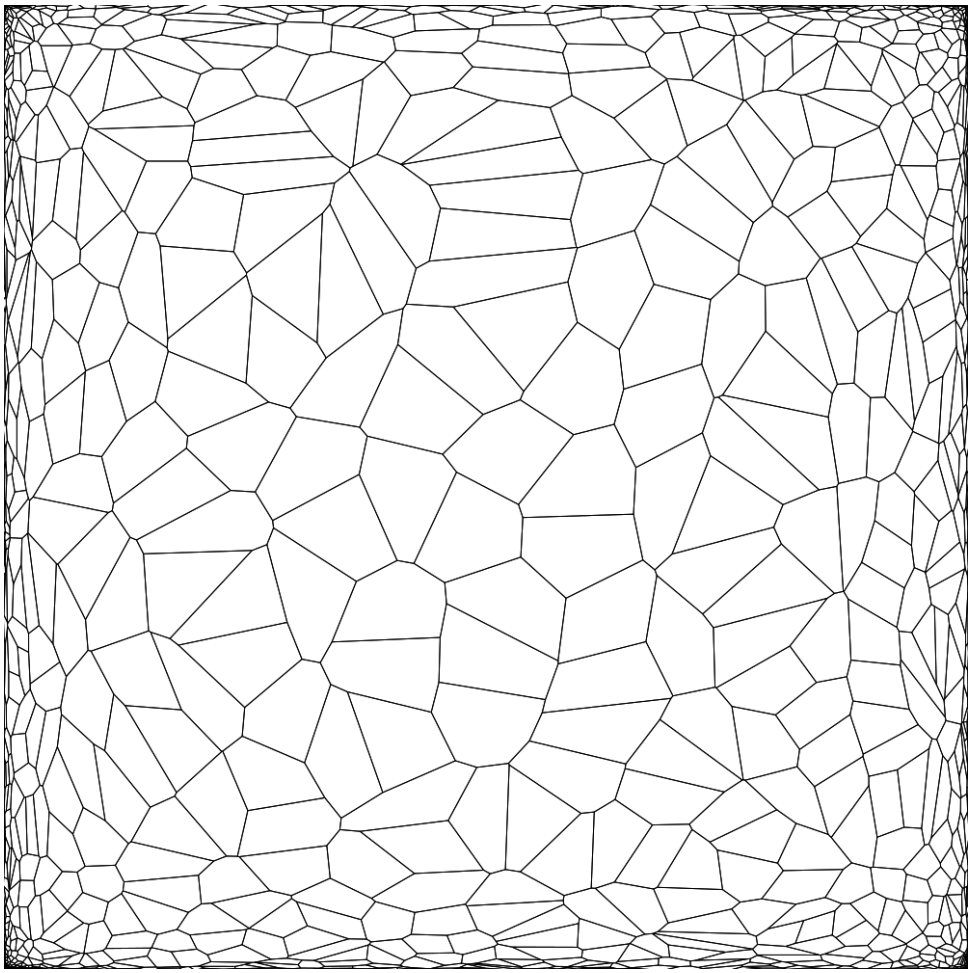


Figure 3: The result of semi-discrete optimal transport with 2,000 sites. The power diagram is optimized so that cell areas correspond to a target density function.

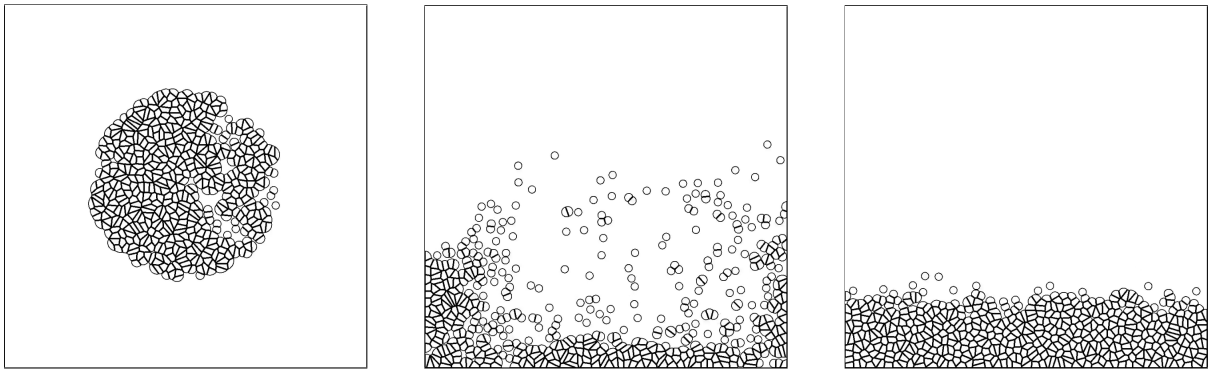


Figure 4: Three frames from the fluid simulation, showing the initial state (left), an intermediate splashing state (middle), and a later, more settled state (right).